

The affine bulk-given-spike law in spiked ReLU networks: exact one-layer analysis, the Bayes knee, and the spike-marginal recursion

One trained case — `experiments/affine_conditional_layer1`

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Abstract

The binned / analytic K=2 predictors compress the bulk-given-spike conditional law of the coordinate-spike model $M_\ell = W_\ell + e_1 e_1^\top$ ($W \sim N(0, 1/n)$, no training) to an *affine family* in the spike coordinate. Empirically this is excellent *post*-activation and mediocre *pre*-activation at depth. This note collects the mechanism, established in three steps with exact computations and MC at widths 128–4096: (1) after *one* layer the post-activation conditional moments are affine with $1 - R^2 \propto 1/n$ and closed-form constants; (2) the failure at the next *pre*-activation is a Bayes-posterior **knee** — the Tweedie correction $\omega^2 \partial_a \log p(a)$ of a rectified (atom + branch) marginal seen through a Gaussian channel — which is $O(1)$ in R^2 terms and width-flat, rank-one across coordinates, and predicted with zero shape parameters; (3) depth propagation is governed by a **one-dimensional recursion on the spike marginal**, $p_{\ell+1} = \varphi_{\omega_\ell} * (c_\ell \text{ReLU} + \mu_\ell)_\# p_\ell$, whose per-layer scalar inputs are measurable, whose density prediction is accurate to $O(n^{-1/2})$ (dominated by one scalar, $\kappa = w^\top m_1$; including it gives $O(n^{-1})$), and which shows the knee is a damped fixed-point object ($q \approx 0.5$ per layer), not an accumulating one.

1 Setup and notation

Layers $Z^{(\ell+1)} = M_{\ell+1} H^{(\ell)}$, $H^{(\ell)} = \text{ReLU}(Z^{(\ell)})$, $H^{(0)} = X \sim N(0, I_n)$, $M_\ell = W_\ell + e_1 e_1^\top$, $W_\ell \sim N(0, 1/n)$ i.i.d. *Spike* $A_\ell = Z_0^{(\ell)}$; *bulk* $Z_1^{(\ell)}$. The surrogates assume, per layer, the affine family

$$Z_{\text{bulk}}^{(\ell)} \mid A_\ell = a \rightsquigarrow N(m_0 + m_1 a, \Sigma_0 + \Sigma_1 a),$$

and we measure its quality by the mass-weighted pooled $1 - R^2$ of weighted-least-squares affine fits of the true conditional moment curves (split-half debiased when the curves come from MC). Throughout φ, Φ are the standard normal pdf/cdf, $\phi_0 = \varphi(0) = 1/\sqrt{2\pi}$, $\tau^2 = \text{Var}(A_1) = \|M_1[0, :]\|^2 \approx 2$.

2 One layer: affine to $O(1/n)$, exactly

Conditional on $A_1 = s$, the bulk pre-activation is *exactly* Gaussian (joint Gaussianity of $Z^{(1)} = M_1 X$):

$$Z_{\text{bulk}}^{(1)} \mid s \sim N(\beta s, C), \quad \beta = (M_1 M_1^\top)_{\text{bulk}, 0} / \tau^2 = O(n^{-1/2}) \text{ per coord}, \quad C \text{ const in } s.$$

So pre-activation is trivially affine, and all post-activation curvature comes from the ReLU moment maps $f(\mu, \sigma) = \mathbb{E} \text{ReLU}(\mu, \sigma^2)$, $g = \text{Var} \text{ReLU}(\mu, \sigma^2)$, evaluated at $\mu = \beta_i s$: analytic maps (the kink is convolved away by the $O(1)$ conditional noise σ_i), probed on an interval of width $O(\beta_i \tau) = O(n^{-1/2})$. The affine model is the tangent line; the residual is the quadratic Taylor term. With $f'(0) = \frac{1}{2}$, $f''(0) = \phi_0 / \sigma$, $g'(0) = \sigma \phi_0$, $g''(0) = \frac{1}{2} - \frac{1}{\pi}$, best-affine fitting under the Gaussian mass of s gives

$$1 - R_{\text{mean}, i}^2 \simeq \frac{2\phi_0^2 \beta_i^2 \tau^2}{\sigma_i^2}, \quad 1 - R_{\text{var}, i}^2 \simeq \left(\frac{1/2 - 1/\pi}{\sigma_i \phi_0} \right)^2 \frac{\beta_i^2 \tau^2}{2} \quad (\text{ratio} \approx 3.07), \quad (1)$$

both $\propto \beta_i^2 \propto 1/n$.

Fact 1 (measured; widths 64–4096, 5 seeds, exact rectified-Gaussian conditional curves, no MC). *Pooled* $1 - R^2$: mean 1.87×10^{-3} , variance 6.05×10^{-4} , off-diagonal covariance 1.07×10^{-3} at $n = 512$; $2.38 \times 10^{-4} / 7.7 \times 10^{-5} / 1.35 \times 10^{-4}$ at $n = 4096$; width exponents $-0.97, -0.96, -1.00$. Per-coordinate collapse onto (1) over nine decades. Fitted slope vector $= \Phi(0)\beta$ to machine precision ($\cos = 1 - 10^{-16}$); only $\cos \approx 1/\sqrt{2}$ of it lies along the naked column $W_{[1:,0]}$ (the other half is row overlap $W_{\text{bulk}} w_0^\top$). The conditional variance is not constant: it tilts with slope $2\phi_0\sigma_i \times$ the mean slope (≈ 0.80 ; measured 0.795–0.797) — a constant- Σ model errs at $O(n^{-1/2})$, the affine family at $O(1/n)$. The residual is a parabola in s : edge cells ($|s| \geq 2.5\tau$) are $3.3\times$ worse, central ($|s| \leq 1.5\tau$) $3.2\times$ better, same $1/n$ law; true failure needs $|\beta_i s| \sim \sigma_i$, i.e. a $\sim \sqrt{2n}$ -sd spike excursion. (Figs. 1–2.)

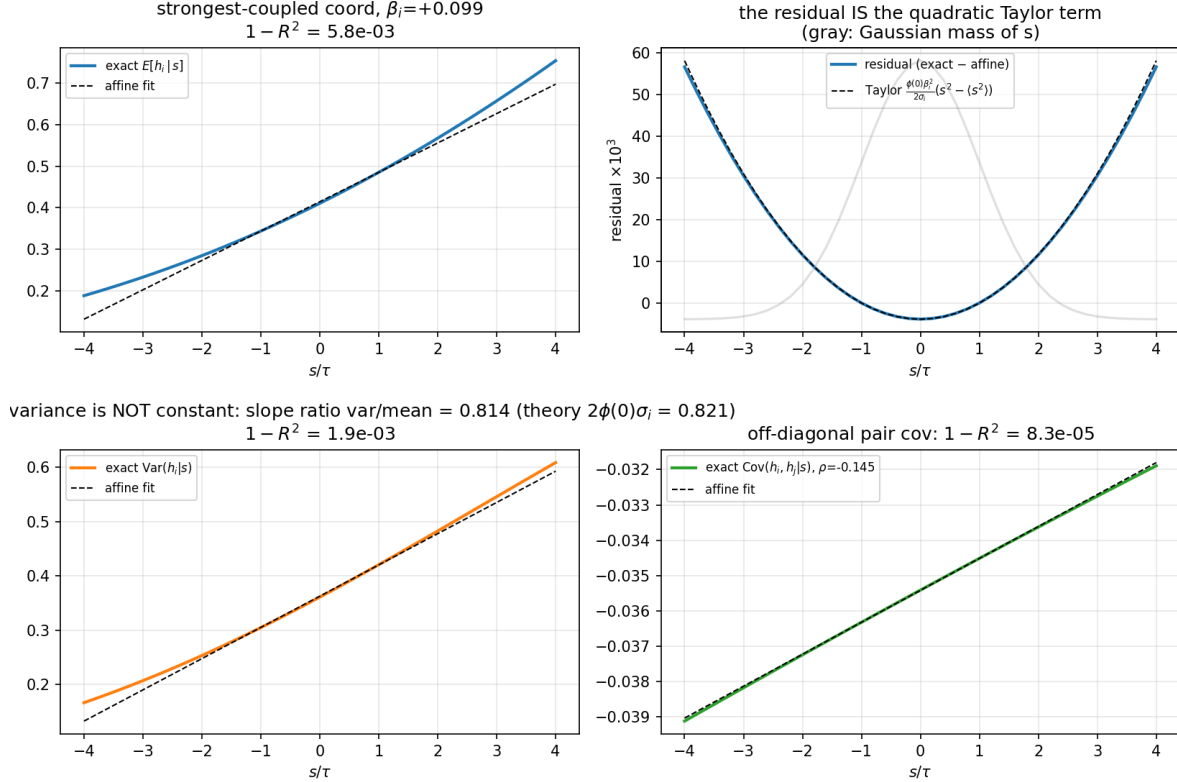


Figure 1: Layer 1, $n = 512$, strongest-coupled coordinate: the affine fit and its residual $\equiv$ the quadratic Taylor term; the conditional variance tilts at $0.8\times$ the mean slope; off-diagonal covariances are straightest.

3 Why post-affine does not give the next pre-affine: the Bayes knee

The next spike is, *exactly*,

$$A_{\ell+1} = c_\ell \text{ReLU}(A_\ell) + \xi_\ell, \quad c_\ell = M_{\ell+1}[0,0] \approx 1, \quad \xi_\ell = w_\ell^\top H_{\text{bulk}}^{(\ell)}, \quad (2)$$

with $\xi_\ell \approx N(\mu_\ell, \omega_\ell^2)$ a fresh CLT sum ($\mu_\ell = w^\top m_0$ is an $O(1)$ random-per-net offset; $\omega_2^2 \approx 0.32$). Conditioning the new bulk on $A_{\ell+1} = a$ routes through the posterior of the old variables: every coordinate obeys

$$\mathbb{E}[Z_i^{(\ell+1)} | a] = \alpha_i + \rho_i a + \lambda_i g_{\ell+1}(a), \quad g_{\ell+1}(a) = \mathbb{E}[\text{ReLU}(A_\ell) | A_{\ell+1} = a],$$

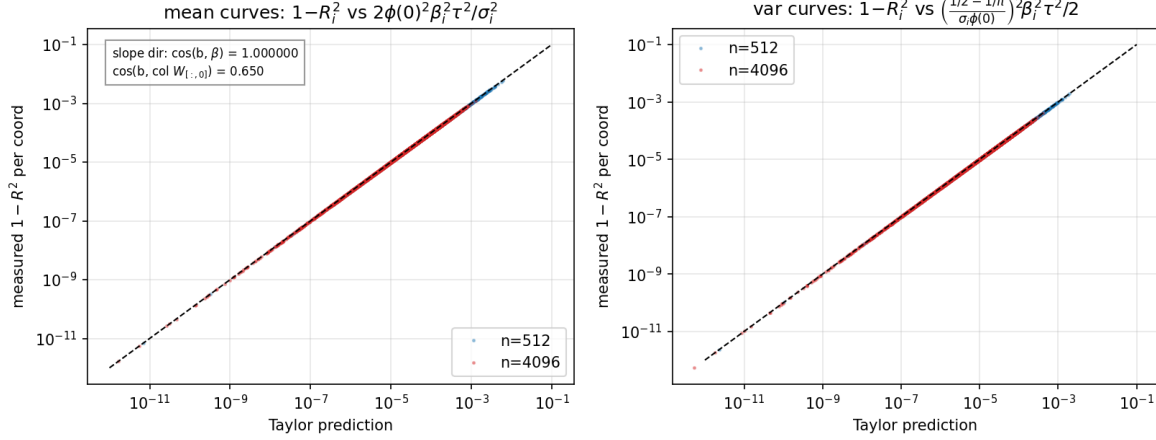


Figure 2: Per-coordinate measured $1 - R^2$ vs. the closed forms (1).

one *shared* profile g with $O(n^{-1/2})$ coordinate amplitudes λ_i . Tweedie's formula for the Gaussian channel (2) gives

$$c g(a) = a - \mu - \omega^2 \partial_a \log p_{A_{\ell+1}}(a) : \quad (3)$$

g is affine iff the score is affine iff $A_{\ell+1}$ is Gaussian (Cramér) — but $\text{ReLU}(A_\ell)$ has an atom plus a branch, so g is a **knee**: a plateau (atom side), a linear branch, and a crossover strip of width $\sim \omega$. Because R^2 is scale-free, the $O(n^{-1/2})$ amplitude λ_i divides out: pre-activation affine $1 - R^2$ is $O(1)$ and *width-flat*. The conditional covariance picks up the rank-one term $\tilde{c}\tilde{c}^\top \text{Var}(A_\ell | a)$, whose coefficient is an S-curve with overshoot ($0 \rightarrow 0.50 \rightarrow \text{floor } (\tau^{-2} + \omega^{-2})^{-1} = 0.40$) — verified against MC along the predicted direction at $n = 2048$ (control direction flat).

Fact 2 (measured). *Depth-2/3 nets, $n \in \{256, 1024\}$, 1M samples, split-half debiased: layer-2 PRE $1 - R^2 = 0.156\text{--}0.168$, identical across widths; cross-half rank-one share 0.97–0.99; measured shared profile matches the zero-parameter knee at $\cos = 0.9986\text{--}0.9996$; amplitude $\text{rms}(\lambda)\sqrt{n} = 0.36\text{--}0.39$. The same knee measured at $n = 2048$ (depth 2): pooled PRE 0.155 vs. 0.159 predicted by the exact-in-surrogate quadrature. (Fig. 3.)*

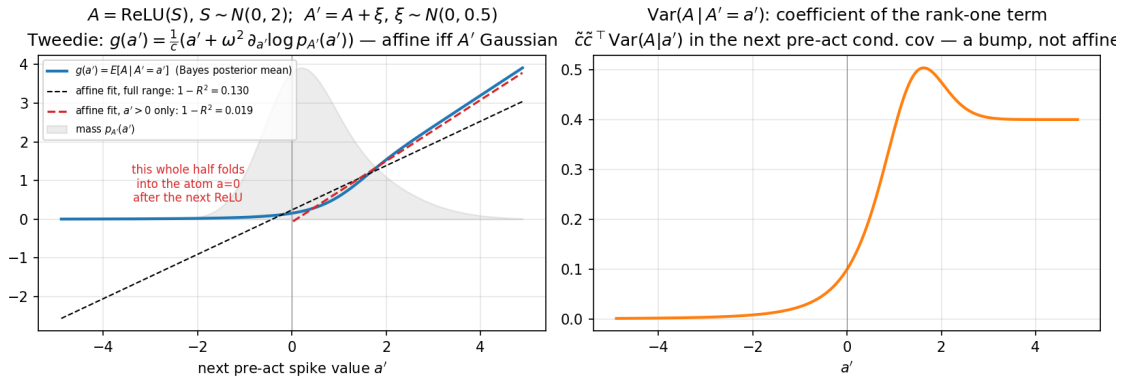


Figure 3: The knee in a toy with the model's scales ($A = \text{ReLU}(S)$, $S \sim N(0, 2)$, $A' = A + \xi$, $\omega^2 = \frac{1}{2}$): posterior mean g (left) and posterior variance (right). Affine fails at $1 - R^2 = 0.130$ on the full range but only 0.019 on $a' > 0$ — the half that survives the next ReLU.

Why the ReLU restores affinity. The ReLU acts *cell-wise* in the conditioning value: post moments at cell a are smooth rectified-Gaussian functions of pre moments at the same a , so it transports the knee

with factor $\Phi(0) = \frac{1}{2}$ and adds only its own $O(1/n)$ curvature (Section 2). What changes is the *conditioning variable*: $a \mapsto \text{ReLU}(a)$ folds the entire $a \leq 0$ half — the atom-dominated plateau and half the crossover — into the single atom point, which the positive-node affine fit does not see. Same curve, two verdicts: toy $0.130 \rightarrow 0.019$; in-model POST (positive bins) = 0.003–0.058 at layers 2–3, i.e. 5–20 $\times$ below PRE, with no depth trend.

4 Depth: the spike-marginal recursion

Iterating (2) marginally,

$$p_{\ell+1}(a) = \int p_{\ell}(y) \varphi_{\omega_{\ell}}(a - c_{\ell} \text{ReLU}(y) - \mu_{\ell}) dy, \quad p_1 = N(0, \tau^2), \quad (4)$$

three measurable scalars per layer. At layer 2 it is closed-form: with $u = a - \mu$, $b = c + \kappa$, $s_-^2 = \kappa^2 \tau^2 + \omega^2$, $s_+^2 = b^2 \tau^2 + \omega^2$ (and $\kappa = w^{\top} m_1$ the small linear pass-through),

$$p_2(a) = \underbrace{\frac{1}{s_-} \varphi\left(\frac{u}{s_-}\right) \Phi\left(-\frac{\kappa \tau u}{\omega s_-}\right)}_{\text{smeared atom } (\rightarrow \frac{1}{2} \varphi_{\omega} \text{ at } \kappa=0)} + \underbrace{\frac{1}{s_+} \varphi\left(\frac{u}{s_+}\right) \Phi\left(\frac{b \tau u}{\omega s_+}\right)}_{\text{skew-normal branch}} \quad (5)$$

verified against quadrature to $\sim 6 \times 10^{-6}$ TV and against 1M-sample MC to TV = 0.004 (the sampling floor; Fig. 4). For $\ell \geq 3$, (4) is the formula.

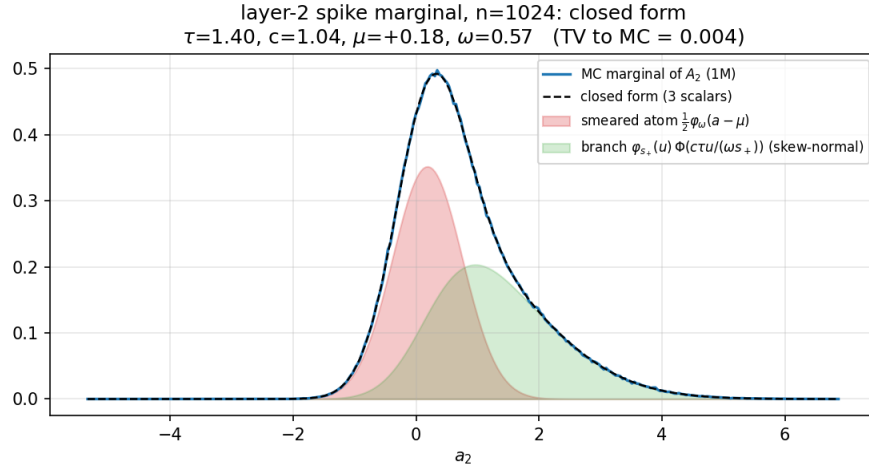


Figure 4: Layer-2 spike marginal at $n = 1024$: closed form (5) vs. MC.

Why the knee passes nicely through layers. (i) The fresh knee at every layer depends on the past *only through the 1-d marginal* (3); it is irrelevant that A_{ℓ} is knee-contaminated and non-Gaussian. (ii) Each layer re-smooths with a *fresh* Gaussian ξ_{ℓ} ; for any $p = q * \varphi_{\omega}$ the posterior-variance identity $\text{Var}(T|a) = \omega^2(1 + \omega^2(\log p)'') \geq 0$ caps the score curvature at $-1/\omega^2$ — the spoon can never sharpen beyond the channel noise, at any depth. (iii) The *old* knee propagates only as $Sg(a) = \mathbb{E}[g_{\ell}(A_{\ell}) \mid A_{\ell+1} = a]$ — smoothed by the posterior kernel (width $\approx$ the Var floor ≈ 0.6 , the knee’s own feature size) and affine-projected: measured damping $q = \|(I - \Pi)Sg\|/\|(I - \Pi)g\| = 0.34\text{--}0.75$ (≈ 0.5 stationary), so nonaffine power saturates geometrically instead of accumulating.

Fact 3 (measured, depth 3). *Marginals: recursion vs. MC, TV = 0.004–0.016 at layers 2 and 3 (all four nets). Layer-3 PRE $1 - R^2 = 0.057\text{--}0.172$: no systematic growth over layer 2; the seed scatter tracks the random offsets μ_{ℓ} (which decide where the knee sits in the mass), and the recursion reproduces the seed*

ordering. Layer-3 profile = recursion knee at $\cos = 0.984\text{--}0.999$; rank-one share $0.88\text{--}0.96$, with the second component correlating with the predicted damped layer-2 knee at $\cos = 0.43\text{--}0.59$. Under a stationary channel ($c = 1, \mu = 0, \omega^2$ fixed) the fresh-knee $1 - R^2$ relaxes $0.103 \rightarrow 0.037$ over six layers with self-similar shape ($\cos \rightarrow 0.997$). Caveat: this unnormalized net contracts bulk variance $\times \sim \frac{1}{3}$ per layer, so ω_ℓ^2 shrinks ($0.32 \rightarrow 0.12$) — very deep behavior is driven by that and by μ -drift, not by knee accumulation. (Figs. 5–7.)

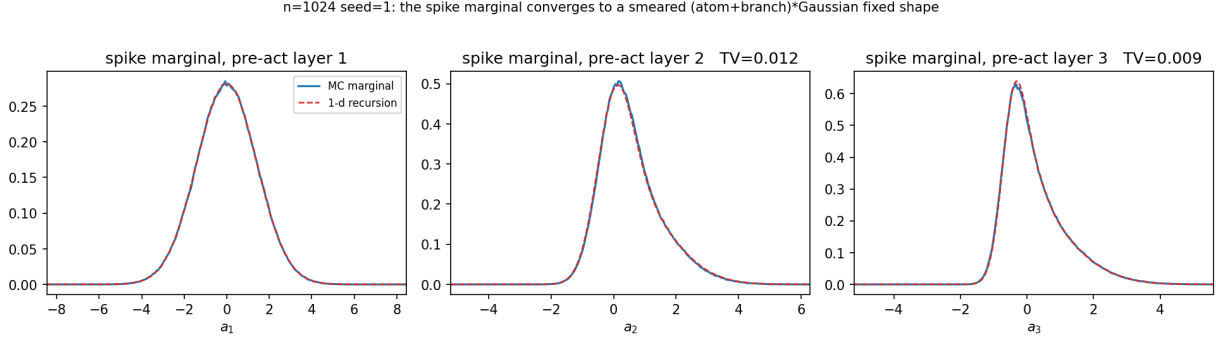


Figure 5: Spike marginals, layers 1–3 ($n = 1024$): MC vs. recursion (4).

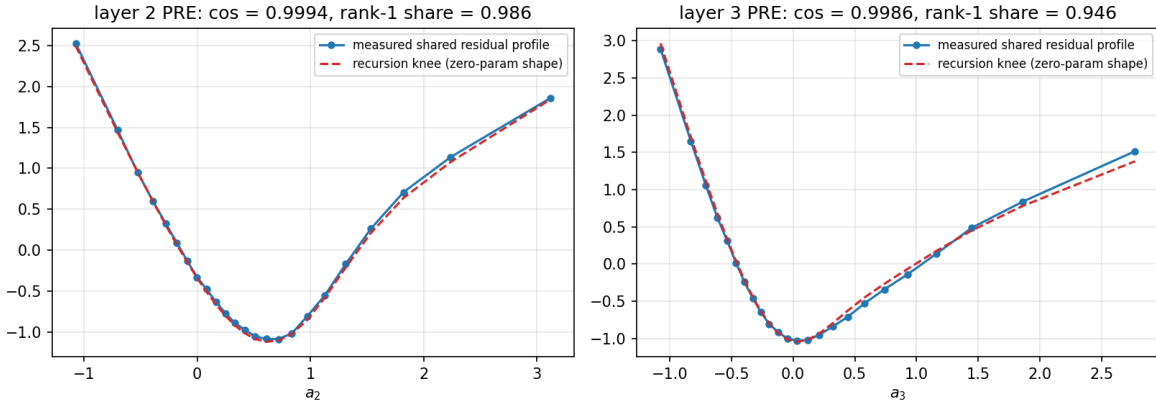


Figure 6: Measured shared affine-residual profiles at layers 2 and 3 vs. the zero-parameter recursion knee.

5 Accuracy of the 1-d reduction: what is dropped and how it scales

The recursion idealizes $\xi | A_\ell \sim N(\mu, \omega^2)$, independent of A_ℓ . The exact conditional channel is

$$\xi | A_\ell = a \sim N\left(\mu + \underbrace{\kappa a}_{\kappa = w^\top m_1} + \underbrace{\lambda_w g_\ell(a)}_{\lambda_w = w^\top \lambda}, \omega^2(1 + O(a/\sqrt{n}))\right) + \text{Edgeworth},$$

so the bulk (“the other directions”) is *not* small — ξ is $O(1)$ and is precisely the smoothing — but everything about it beyond $(\mu, \omega^2, \text{Gaussian})$ is $O(n^{-1/2})$ or smaller:

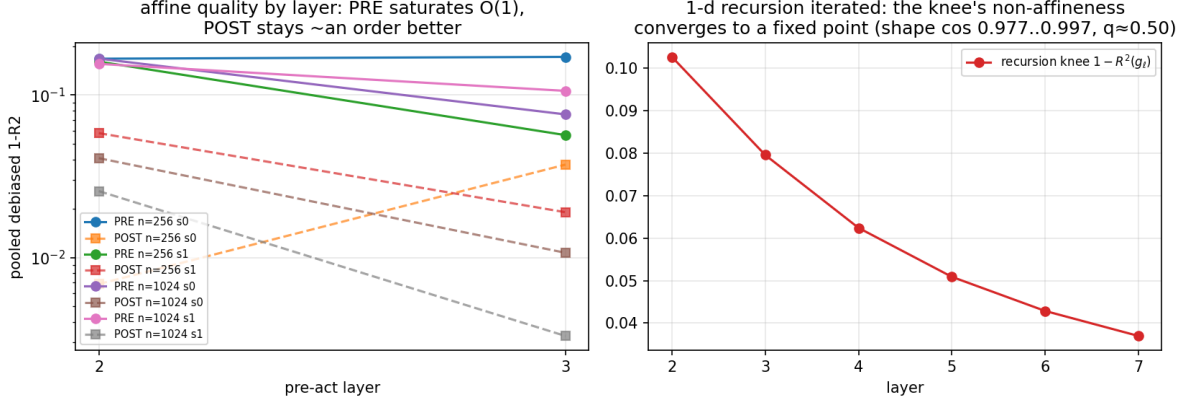


Figure 7: Left: PRE saturates at $O(1)$, POST stays an order better, both without depth growth. Right: stationary-channel iteration of (4).

dropped term	size (theory)	measured	note
mean tilt $\kappa = w^\top m_1$	$N(0, \ m_1\ ^2/n)$, $\ m_1\ = \sqrt{1/8} \approx 0.354$	$ \kappa \sqrt{n} \approx 0.35$	dominant
old-knee leak $\lambda_w g_\ell(a)$	$O(n^{-1/2})$	inside $E[\xi a]$ dev.	subdominant
variance tilt $\text{Var}(\xi a)/\omega^2 - 1$	$O(n^{-1/2})$	$\lesssim 0.03$ at $n=128$, noise-floor beyond	tiny
skewness of ξ	$O(n^{-1})$ (random-sign w)	$0.08 \rightarrow 0.001$, $128 \rightarrow 2048$	negligible

Hence the predicted *density* error amplitude is $O(n^{-1/2})$, i.e. $D^2 = \int (p - \hat{p}_{\text{rec}})^2 \propto 1/n$, and including the single measured scalar κ should advance one order.

Fact 4 (measured; cross-half-debiased D^2 , widths 128–2048, 2 seeds). *Layer 3: $D^2 \propto n^{-1.00}$ (3-scalar recursion) and $\propto n^{-1.8}$ with the κ tilt included. Layer 2 with κ : D^2 is zero within the 1M-sample noise floor ($|D^2| \lesssim 10^{-5}$, $TV \approx 0.004$ – 0.008). Per seed, the 3-scalar error is $\approx \kappa^2 \text{Var}(a)$ — one scalar owns the leading error. (Fig. 8.)*

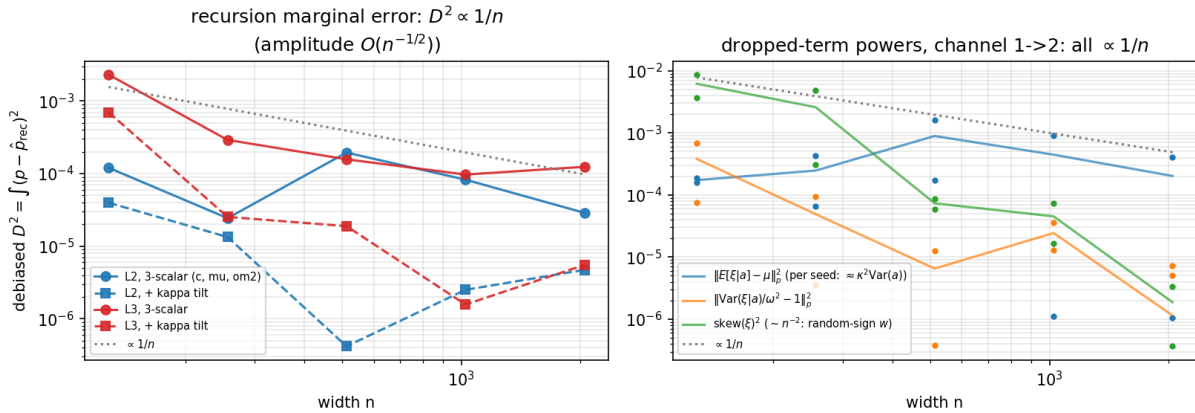


Figure 8: Left: debiased density error of the recursion vs. width. Right: dropped-term powers.

6 Consequences and open ends

- **Why affine-K2 predictors work:** the entire conditional structure is affine up to (a) $O(1/n)$ within a layer (Section 2) and (b) one rank-one knee profile whose WLS-affine projection is zero by construction ($g \perp \{1, a\}$), so it leaks into propagated *moments* only through the next ReLU’s curvature, $O(1/n)$ per coordinate — consistent with the observed n^{-2} MSE law and analytic=binned parity.
- **Cheap upgrade:** augment the affine family with the single basis function $g_\ell(a)$ from (4)–(3) (rank-one knee correction; 3 scalars per layer buy the profile).
- **Discriminator:** this mechanism predicts deep POST $1 - R^2$ floors at an n -independent value set by the knee’s positive shoulder rather than falling like $1/n$ — checkable against the `r2v2_*` rows in `checkpoints/analytic_kprop/r2/`.
- **Caveat:** the unnormalized $W \sim N(0, 1/n)$ net contracts bulk variance per layer; with He-type $2/n$ scaling the channel scalars would be depth-stationary and the fixed-point picture of Section 4 applies verbatim.

Artifacts. Notebook + builder: `experiments/affine_conditional_layer1/` (layer-1 exact study); scripts `knee_theory`, `knee_mc`, `knee_analysis` (layer-2 knee at scale), `knee_depth_mc`, `knee_depth_analysis` (depth 2–3), `knee_channel_mc`, `knee_channel_analysis` (channel structure, recursion accuracy). Caches: `checkpoints/affine_con` (`aff_l1_*`, `knee2_*`, `knee3_*`, `chanmc_*`, `figs/`). Shared kernel added: `Mecha_preds/_utils.py::exact_relu_covariance_pa`